

Michelle Elias

M.Sc. Biophysics Student

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Computational biologist specialising in physiologically based pharmacokinetic modelling, digital twins, and reproducible open-source science. Interested in mechanistic modelling approaches that bridge systems biology and precision medicine.

Education

- 10/2025
– present
- M.Sc. Biophysics**
Humboldt University of Berlin, Germany
Thesis: *in progress*
Selected courses: Systems Biology: Computational Analysis and Interpretation of High-throughput Data; Systems Biology: Dynamic Modeling and Data Integration; Molecular, Cellular and Medical Biophysics
Course project: *Investigating Effects of Polysome Profiling Resolution for a Translation Reporter Assay with Deep Learning* — CNN-based prediction of mean ribosome load (MRL) from 5'UTR sequences using 1D convolutional neural networks; evaluation across multiple polysome resolution schemes with 5-fold cross-validation; model interpretation via Integrated Gradients (Captum).
🔗 [Code](#)
- 10/2019
– 04/2025
- B.Sc. Biology**
Humboldt University of Berlin, Germany
Final grade: 1.9
Thesis: *“A Physiologically-Based Pharmacokinetic (PBPK) Model of the Sulfonyleurea Glimepiride”*
Supervisor: Dr. Matthias König, Systems Medicine of the Liver, Humboldt University of Berlin
Bachelor Project grade: 1.0 | Thesis grade: 1.0
Development of a whole-body PBPK model of glimepiride based on curated clinical data, investigating pharmacokinetic variability due to renal and hepatic impairment, bodyweight, and CYP2C9 genetic variants. Validated against multiple clinical datasets and published as a digital twin model (published as: *A Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment*).
- 2013
– 2019
- Abitur**
Georg-Büchner-Gymnasium, Berlin
Bilingual school (English/German) with a focus on natural sciences.
Selected advanced courses: Mathematics and Arts

Research Experience

- 01/2026
– present
- Student Assistant**
Systems Medicine of the Liver (König Lab), Institute for Theoretical Biology, Humboldt University of Berlin
- Contributing to the DFG-funded priority programme **SPP2311**; developing **FEMVis**, a web platform for FEM simulation visualization.
 - Coordinated cross-programme data collection for SPP2311 by designing and distributing project and model survey forms to all subproject partners.
 - Supporting scientific manuscript development and editing for ongoing publications in the group.
 - Working on reproducible modelling workflows, tool usability, and model interpretation.
- 04/2024
– 06/2024
- Research Intern**
Cardiovascular Precision Medicines, Bayer AG, Wuppertal
- Cultured hiPSC-derived cardiomyocytes and performed gene knockdown experiments via siRNA transfection.
 - Conducted molecular analyses including qPCR, Western blot, and DNA/RNA/protein extractions.
 - Performed statistical analysis and data visualization of experimental results in support of the research team.
 - Received an excellent internship reference.

Publications

First-author publications

- **A Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment**
Elias M and König M. *Frontiers in Pharmacology*, 2025. (*published*)
doi:10.3389/fphar.2025.1686415 | Preprint |  Code |  Web App
- **Reproducibility of a Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment**
Elias M and König M. *Physiome*, 2025. (*published*)
doi:10.36903/physiome.28379193 |  Code
- **Reproducibility of a Physiologically Based Pharmacokinetic and Pharmacodynamic (PBPK/PD) Model of Dapagliflozin**
Elias M, Myshkina M, Nemitz N, and König M. *Physiome*, 2026. (*accepted*)
doi:10.36903/physiome.31368556 |  Code
- **Reproducibility of a Digital Twin of Canagliflozin Pharmacokinetics and Pharmacodynamics in Type 2 Diabetes Mellitus**
Elias M, Myshkina M, Tereshchuk V, and König M. *Physiome*, 2026. (*submitted*)
 Code
- **Reproducibility of a Digital Twin of Empagliflozin Pharmacokinetics and Pharmacodynamics in Type 2 Diabetes Mellitus**
Elias M, Myshkina M, Alejandro J, Babaeva M, and König M. *Physiome*, 2026. (*in preparation*)
 Code

Co-authored publications

- **A Physiologically Based Pharmacokinetic and Pharmacodynamic (PBPK/PD) Model of Dapagliflozin in Type 2 Diabetes Mellitus: The Effect of Dosing, Hepatorenal Impairment, and Food**
Nemitz N, **Elias M**, and König M. *Pharmaceutics*, 2026. (*published*)
doi:10.3390/pharmaceutics18030287 | Preprint |  Code
- **Reproducibility of a Digital Twin of the Angiotensin II Receptor Blocker Losartan**
Myshkina M, **Elias M**, Tensil E, and König M. *Physiome*, 2026. (*accepted*)
doi:10.36903/physiome.31359823 |  Code
- **Rivaroxaban Pharmacokinetics and Coagulation Effects: Reproducibility of a Physiologically Based Modeling Approach**
Myshkina M, **Elias M**, Casabianca E, and König M. *Physiome*, 2026. (*submitted*)
 Code
- **A Digital Twin of Canagliflozin Pharmacokinetics and Pharmacodynamics in Type 2 Diabetes Mellitus**
Tereshchuk V, **Elias M**, and König M. *Preprints*, 2026. (*submitted*)
preprints.org/202601.2095 |  Code
- **A Digital Twin of Empagliflozin Pharmacokinetics and Pharmacodynamics in Type 2 Diabetes Mellitus**
Alejandro J, **Elias M**, Babaeva M, and König M. *Journal TBD*, 2026. (*in preparation*)
 Code

Work Experience

Healthcare

- | | |
|----------------------|---|
| 04/2025
– 12/2025 | Student Assistant, Medical Controlling
<i>Martin Luther Krankenhaus, Berlin</i>
Supporting digitalization of patient records; trained and onboarded new student assistants. |
| 09/2024
– 12/2025 | Student Assistant, Hospital Archive
<i>Ev. Elisabeth Klinik, Berlin</i>
Coordinated archive relocation project including workflow planning and logistics; digitized patient records; trained and onboarded new student assistants. |
| 01/2023
– 12/2023 | Student Assistant, Hospital Administration
<i>Martin Luther Krankenhaus, Berlin</i>
General administrative support including patient file management and insurance processing. Excellent work reference received. |

Social & Humanitarian Work

2022	Student Assistant <i>Johanniter – Ukrainian Refugee Center, Berlin</i> Support for refugee registration, orientation, and connection to essential services.
2021	Student Assistant <i>Malteser – COVID-19 Vaccination Center, Berlin</i> Assisted medical personnel with vaccine administration and patient flow coordination.

Skills

Languages	German (native), Polish (native comprehension), English (C1, TOEFL iBT 113)
Modeling & Systems Biology	PBPK/PD Modeling , Systems Biology, Digital Twins, Multiscale & ODE-based Simulation, Population Simulation, Pharmacogenomics (CYP2C9), Precision Medicine
Data Science & AI	Machine Learning & AI (PyTorch, Captum), CNN-based Sequence Modeling, Parameter Estimation, Computational Simulation
Scientific Workflows	Clinical Data Curation (PK-DB), Systematic Literature Review, FAIR Data Management, Reproducible Workflows
Programming	Python (NumPy, SciPy, pandas, matplotlib, seaborn, PyTorch), R, Linux, HTML/CSS, SBML/libSBML, sbmlsim , sbmlutils , libRoadRunner
Tools & Software	git , PyCharm, VS Code, Jupyter, Marimo, Docker, Google Colab, L^AT_EX , Typst, Overleaf, Excel (Microsoft Certified), Word, PowerPoint, BioRender, Canva, WebPlotDigitizer
Open Science	SBML , OMEX , COMBINE , Open-source publication (GitHub, Physiome), Interactive PK simulation tools & web apps
Laboratory	qPCR, Western blot, siRNA transfection, cell culture (hiPSC-derived cardiomyocytes)